

Planning, Learning and Intelligent Decision Making

Homework 1

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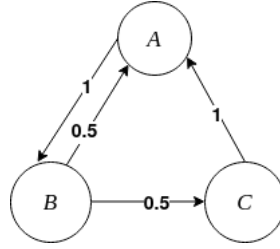
Exercise 1.

(a)

A Markov chain model can be represented as a tuple, (χ, P) , where χ are the states and P the corresponding transition probability matrix. The states for this model are the web pages, $\chi = \{A, B, C\}$. Since, in this model, the bot will select uniformly one of the links in that document, the probabilities of each chosen link are equiprobable. Thus, the transition probability matrix is as follows:

$$P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

The transition diagram of this Markov Chain:



(b)

Let T_{xy} be the average number of steps the bot takes to return to document x when starting in y, so T_{xx} is the average number of steps that the bot takes to return to document x when starting in x. For each path, we need to account for the probabilities associated with each path, P_{xy} .

- T_{AA} :
 $T_{AA} = P_{AB}T_{AB} + P_{BA}T_{BA} + P_{BC}(T_{BC} + P_{CA}T_{CA}) \Leftrightarrow T_{AA} = T_{AB} + 0.5T_{BA} + 0.5(T_{BC} + T_{CA}) \Leftrightarrow T_{AA} = 1 + 0.5 * 1 + 0.5 * 2 \Leftrightarrow T_{AA} = 2.5$
- T_{BB} :
 $T_{BB} = P_{BA}(T_{BA} + P_{AB}T_{AB}) + P_{BC}(T_{BC} + T_{CA} + T_{AB}) \Leftrightarrow T_{BB} = 0.5(T_{BA} + T_{AB}) + 0.5(T_{BC} + T_{CA} + T_{AB}) \Leftrightarrow T_{BB} = 0.5 * 2 + 0.5 * 3 \Leftrightarrow T_{BB} = 2.5$
- T_{CC} :

$$\begin{cases} T_{CC} = P_{CA}T_{CA} + P_{AC}T_{AC} \\ T_{AC} = P_{AB}T_{AB} + P_{BC}T_{BC} \\ T_{BC} = P_{BC}T_{BC} + P_{BA}(T_{BA} + T_{AC}) \end{cases} \Leftrightarrow \begin{cases} T_{CC} = T_{CA} + T_{AC} \\ T_{AC} = T_{AB} + T_{BC} \\ T_{BC} = 0.5T_{BC} + 0.5(T_{BA} + T_{AC}) \end{cases}$$

$$\Leftrightarrow \begin{cases} T_{CC} = 1 + T_{AC} \\ T_{AC} = 1 + T_{BC} \\ T_{BC} = 0.5 + 0.5(T_{BA} + 1 + T_{BC}) \end{cases} \Leftrightarrow \begin{cases} T_{CC} = 1 + T_{AC} \\ T_{AC} = 1 + T_{BC} \\ T_{BC} = 0.5 + 0.5(1 + 1 + T_{BC}) \end{cases}$$

$$\begin{aligned}
&\Leftrightarrow \begin{cases} T_{CC} = 1 + T_{AC} \\ T_{AC} = 1 + T_{BC} \\ T_{BC} = 0.5 + 1 + 0.5T_{BC} \end{cases} &\Leftrightarrow \begin{cases} T_{CC} = 1 + T_{AC} \\ T_{AC} = 1 + T_{BC} \\ 0.5T_{BC} = 1.5 \end{cases} \\
&\Leftrightarrow \begin{cases} T_{CC} = 1 + T_{AC} \\ T_{AC} = 1 + 3 \\ T_{BC} = 3 \end{cases} &\Leftrightarrow \begin{cases} T_{CC} = 5 \\ T_{AC} = 4 \\ T_{BC} = 3 \end{cases}
\end{aligned}$$

(c)

Let μ_x represent the percentage of time in document x , and T_{xx} be the average number of steps to return to document x . For this model, $\mu_x \propto \frac{1}{T_{xx}}$.

$$\mu_A = \frac{1}{T_{AA}} = \frac{1}{\frac{5}{2}} = \frac{2}{5}, \mu_B = \frac{1}{T_{BB}} = \frac{1}{\frac{5}{2}} = \frac{2}{5}, \mu_C = \frac{1}{T_{CC}} = \frac{1}{5}$$

A distribution is considered invariant for the chain, if $\mu = \mu P$, where μ is a row vector whose entries are probabilities summing to 1 and P the transition probability matrix. Considering μ as $[\mu_A \quad \mu_B \quad \mu_C]^T = [\frac{2}{5}, \frac{2}{5}, \frac{1}{5}]^T$.

$$\mu P = [\frac{2}{5}, \frac{2}{5}, \frac{1}{5}]^T \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \end{bmatrix} = [\frac{2}{5}, \frac{2}{5}, \frac{1}{5}]^T = \mu$$

Since $\mu P = \mu$, the distribution is invariant for the chain.